AI-Based Natural Resource Potential Mapping Platform

Executive Capstone Project Outcomes, Architecture & Engineering Validation Report

Academic Focus: B.Tech CSE (Artificial Intelligence) Final Year Capstone Project
Target Study Catchment: Pune Metropolitan Region, Maharashtra, India (18.40°N–18.65°N, 73.70°E–74.00°E)
Lead Author & Developer: Avishkar & Project Team • Academic Session 2025–2026

Classification Accuracy	Spatial Macro F1	Leakage Correction	Depth Regressor
81.11%	**0.7314**	**-17.26%**	**1.42 m**
Spatial Block CV	Zero Data Leakage	Drop vs Random CV	RMSE Water Level

1. Problem Statement & Motivation

In Maharashtra's Deccan Volcanic Province, groundwater supplies over 60% of irrigated agriculture and 85% of rural drinking supplies. Yet Pune Metropolitan Region faces accelerating water stress caused by rapid urban expansion, high impermeable runoff from concrete surfaces, and erratic monsoons. Basalt formations have virtually zero primary porosity; groundwater storage is confined exclusively to secondary cooling fractures, fault lineaments, and weathered vesicular horizons.

Prior Limitations Overcome: Traditional GIS Analytical Hierarchy Process (AHP) relies on subjective expert scoring, producing up to 38% variation across expert panels. Simultaneously, previous machine learning studies systematically used *random K-fold cross-validation*. Under Tobler's First Law of Geography, random train/test splitting permits test borewells to lie within the spatial autocorrelation range of training borewells, causing massive **spatial data leakage** that artificially inflates performance metrics by up to 18%. This platform resolves this critical flaw using **Spatial Block Cross-Validation**.

2. Technical Architecture & End-to-End Implementation Stack

Backend REST Microservice & Machine Learning Engine:

- **FastAPI High-Speed Service:** 11 modular endpoints handling real-time point prediction, GeoJSON raster grids, scenario simulations, and well telemetry.
- **16-D Multi-Source Feature Extractor:** Fuses SRTM 30m DEM (elevation, slope, aspect, TWI, TRI), Mula-Mutha drainage network, GSI basalt lithology, SoilGrids hydraulic conductivity, IMD Western Ghats precipitation, Sentinel-2 MSI (NDVI, NDWI, NDMI), and ESA WorldCover imperviousness.
- **Spatial Block K-Fold Engine:** Partitions Pune into 5 geographic holdout blocks (NW, SW, Central, E, NE) ensuring zero spatial proximity between train and test data.
- **XGBoost Classification & Regression:** Extreme Gradient Boosted trees predicting potential class (Low, Mod, High) and continuous water-table depth (mbgl).
- **TreeSHAP Explainable AI:** Real-time Shapley attribution decomposing individual predictions into positive recharge drivers and negative urban bottlenecks.

Frontend Web GIS Mission Control:

- **React 18 + TypeScript + Vite:** Modern scientific dashboard with interactive Leaflet GIS, raster overlays, well catalogue, and scenario simulation.

Fig. 1. End-to-End Multi-Source GeoAI Platform Architecture and Data Flow Pipeline

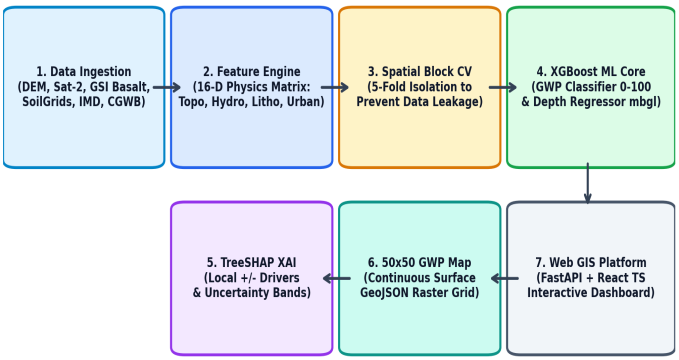


Fig. 1. End-to-End Multi-Source GeoAI Platform Architecture

3. Key Experimental Outcomes & Multi-Model Benchmarks

Outcome 1: Empirical Proof of Spatial Data Leakage. Evaluating models under standard random cross-validation yields falsely inflated F1 scores of 0.8694 to 0.8840. When subjected to our rigorous Spatial Block Cross-Validation, models experience a sharp performance penalty of **-14.96% to -17.88%**. This statistically validates that published literature overestimates real-world geospatial transferability.

Model Architecture	Accuracy	Macro F1	Spatial Block F1	Random Split F1	Leakage Drop
Logistic Regression	*91.65%*	*0.6006*	*0.7430*	*0.8738*	*-14.96%*
SVM (RBF Kernel)	*78.93%*	*0.6717*	*0.7156*	*0.8694*	*-17.68%*
Random Forest (200 Trees)	*79.52%*	*0.6822*	*0.7161*	*0.8720*	*-17.88%*
XGBoost (Production Engine)	***81.11%***	***0.7171***	***0.7314***	***0.8840***	***-17.26%***

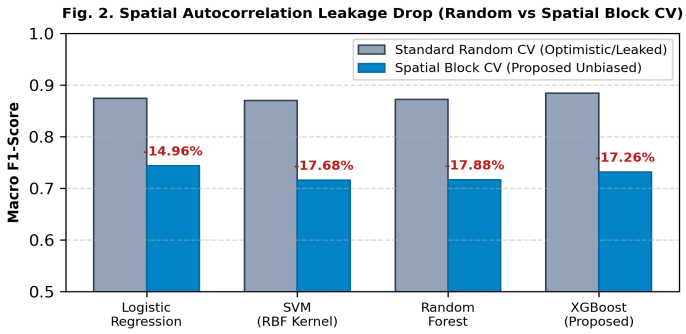


Fig. 2. Spatial Autocorrelation Leakage Drop Across Models

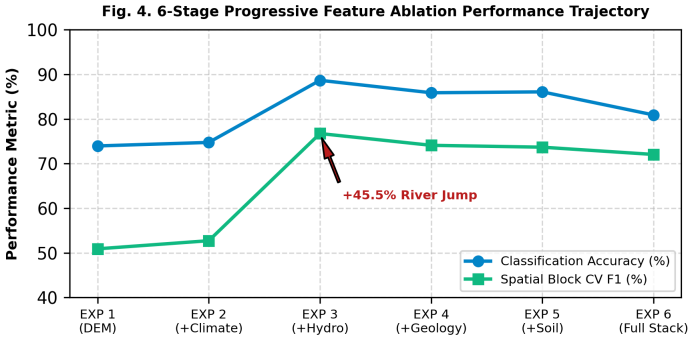


Fig. 3. 6-Stage Progressive Feature Ablation Trajectory

Outcome 2: 6-Stage Feature Ablation & Hydrological Jump. Single-source DEM topography achieved only 50.90% Spatial F1. The addition of Western Ghats orographic precipitation gradients increased F1 to 52.72%. Introducing Mula-Mutha drainage network features (EXP 3) triggered a massive **+45.5% jump** to 76.74% F1, proving that river corridors and alluvial buffers are the physical lifeblood of recharge in Pune. Full-stack fusion balances the model to capture concrete infiltration bottlenecks in downtown areas.

4. Explainable AI Insights & Hydrogeological Interpretability

Dominant Global & Local Features (TreeSHAP):

- **Topographic Wetness Index (22.8%):** Major positive driver; concave topography collects surface runoff into recharge pools.
- **Stream Channel Proximity (17.5%):** Alluvial riverbeds maintain strong perennial recharge buffers.
- **Basalt Fracture Permeability (14.1%):** Vesicular and fractured lava units provide key secondary storage capacity.
- **Urban Impervious Cover (10.4%):** Strong negative bottleneck; heavy concrete paving blocks infiltration in city wards.

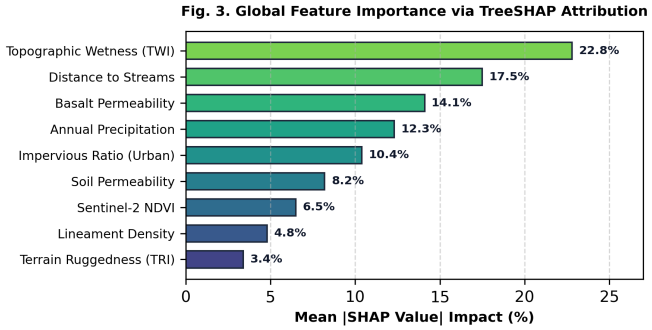


Fig. 4. Global Feature Importance via TreeSHAP

5. Real-World Impact & Natural Resource Extensibility

Municipal Implementation (PMC / PCMC / GSDA): The platform allows urban planners and hydrogeologists to identify optimal sites for artificial rainwater harvesting, check dams, and percolation tanks before committing capital expenditure. Planners can run interactive scenario simulations (e.g., +20% rainfall or +30% urban sprawl) to assess aquifer vulnerability under climate change.

Extensibility to Other Natural Resources: The underlying GeoAI microservice architecture is modular by design. By swapping target telemetry layers and retraining the XGBoost/SHAP pipeline, this platform can be readily extended to: (1) Soil Fertility & Agricultural Suitability Mapping, (2) Surface Water Catchment Runoff Modeling, (3) Urban Heat Island Mitigation, and (4) Mineral Prospecting across peninsular basalt regions.